

Claude Statistical Test Evaluation

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Prompt Used

“I have a study with X patients in the treatment arm, X patients in the control arm, X responders in the treatment group, and X responders in the control group. Please use the attached file to run the appropriate statistical test and provide summary statistics.”

Multiple Codes

Chat

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

Test	Correctness	Time	Format	Uses Our Code
test(10,NA,1.2,3) invalid	T (pointed out NA and 1.2 which will cause error, correct decision)	11 sec	Clean, simple	Y
test(50,40,30,10) chisq, dependent	T (chi sq, test statistic=11.03, p < 0.001. Correctly used p-value labels, correct decision)	26 sec	Clean, with table and sections	N (R not available, but Claude followed my logic)
test(50,40,26,19) chisq, independent	T (chi sq, test statistic=0.18, p=0.67. Correctly used p-value labels, correct decision)	25 sec	Clean, with table and sections	N (R not available, but Claude followed my logic)
test(20,10,5,2) fisher, independent	T (Fisher, OR=1.33 (small rounding deviation), p=1.0. Correctly used p-value labels, correct decision)	27 sec	Clean, simple, with a few sections	N (R not available, but Claude followed my logic)

test(10,15,10,1) fisher, dependent	T (Fisher, OR not shown, p=3.37e-6. Correctly used p-value labels, correct decision)	38 sec	Clean, with sections and table	N (R not available, but Claude followed my logic)
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Code

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
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```

Test	Correctness	Time	Format	Uses Our Code
test(10,NA,1.2,3) invalid	T (correctly specified missing and invalid input, give suggestions)	19 sec	Simple format	Y (using our logic)
test(50,40,30,10) chisq, dependent	T (correct p value, test statistics, and decision)	15 sec	Simple, with section and table	Y
test(50,40,26,19) chisq, independent	T (correct p value, test statistics, and decision, chisq)	15 sec	Simple and brief with table	Y
test(20,10,5,2) fisher, independent	T (correct odds ratio, test statistics, and decision, fisher)	15 sec	Short, brief, with table	Y
test(10,15,10,1) fisher, dependent	T (correct odds ratio, test statistics, and decision, fisher)	18 sec	Short, brief, with summary table	Y

Single Code

Parameters: *n.tx*, *n.ctrl*, *resp.tx*, *resp.ctrl*

Chat

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
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```

Test	Accuracy	Time	Results Format	Uses Our Code
test(10,NA,1.2,3) invalid	T (flags NA and 1.2 errors and asks for correct inputs)	20 sec	Brief, clear explanation of errors and offers correction guidelines	Y (to acknowledge errors via our standards)
test(50,40,30,10) chisq, dependent	F (incorrect chisq statistic and p-value, although correct significant conclusion)	39 sec	Test selection, contingency table, exact hypothesis test output as coded, summary stats, RR/OR interpretation	N (R not available, manually replicated in Python)
test(50,40,26,19) chisq, independent	T	43 sec	Study inputs, test selection, contingency table, expected counts, test results, exact conclusion as coded, in plain terms	N (R not available, manually executed)

test(20,10,5,2) fisher, independent	F (uses Fisher's and gets correct p-value but incorrect OR)	1 min 2 sec	Study inputs, test selection, test output, summary statistics, conclusion (not as coded), interpretation	N (R not available, manually executed)
test(10,15,10,1) fisher, dependent	T	46 sec	Study inputs, test selection, summary statistics, conclusion (not as coded), note (small sample size)	N (R not available, manually executed)

Code

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

Test	Accuracy	Time	Results Format	Uses Our Code
test(10,NA,1.2,3) invalid	T	10 sec	Detects errors based on code and returns error message	Y
test(50,40,30,10) chisq, dependent	T	16 sec	Contingency table, statistical test, summary statistics, conclusion (not as coded)	Y
test(50,40,26,19) chisq, independent	T	31 sec	Contingency table, statistical test, summary statistics, conclusion (not as coded)	Y
test(20,10,5,2) fisher, independent	T	33 sec	Contingency table, statistical test, summary statistics, conclusion (not as coded)	Y
test(10,15,10,1) fisher, dependent	T	16 sec	Contingency table, statistical test, summary statistics, conclusion (not as coded)	Y

Conclusion

Condition	Mean Execution Time	Correctness
Multiple codes, Chat	25.4 sec	5/5
Multiple codes, Code	16.4 sec	5/5
Single code, Chat	42 sec	3/5
Single code, Code	21.2 sec	5/5

We can observe clearly that **multiple codes are more efficient than a single code chunk**, and **Code executes more efficiently compared to Chat**. Chat also has a higher mistake rate when using single code. With multiple codes, the summary Claude provides is usually shorter and more brief compared to those generated using single code.

As a result, using multiple codes is more efficient than using single code, and using Code is more efficient and accurate than Chat.